$$\text{binomial therom: } (x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

let $a=1$, 1^n always =1 so all x terms are just multiplied by 1

$$(x+1)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b 1\right]$$

let $x' = x-1$ to cancel the $+1$ and produce x^r

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-1)^b\right]$$

bring out the $(x-1)^r$ to get recurssion (by applying the identity to that term)

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-1)^b\right] + (x-1)^r$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-1)^b\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2)^b\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-3)^b\right] + (x-3)^r$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-1)^b\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2)^b\right] + ... + \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-x)^b\right] + (x-x)^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

apply the identity to the b^c terms assymetrically to get sums of x^0

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{0}b^0 + \binom{r}{1}b^1 + \binom{r}{2}b^2 + ... + \binom{r}{r-1}b^{r-1}\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}b^0 + \binom{r}{1}\sum_{c=0}^{b-1}\left[\binom{1}{0}c^0\right] + \binom{r}{2}\sum_{c=0}^{b-1}\left[\binom{2}{0}c^0 + \binom{2}{1}\sum_{d=0}^{c-1}\left[\binom{1}{0}d^0\right]\right] + ... + \binom{r}{r-1}b^{r-1}\right] + 0^r$$

$$x^r = \binom{r}{0}\sum_{b=0}^{x-1}[b^0] + \binom{r}{1}\binom{1}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + \binom{r}{2}\binom{2}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + \binom{r}{2}\binom{2}{1}\binom{1}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[d^0] + ... + \binom{r}{r-1}b^{r-1}] + 0^r$$

$$x^r = \binom{r}{0}\sum_{b=0}^{x-1}[b^0] + \left(\binom{r}{1}\binom{1}{0} + \binom{r}{2}\binom{2}{0}\right)\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + \binom{r}{2}\binom{2}{1}\binom{1}{0}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[d^0] + ... + \binom{r}{r-1}b^{r-1}] + 0^r$$

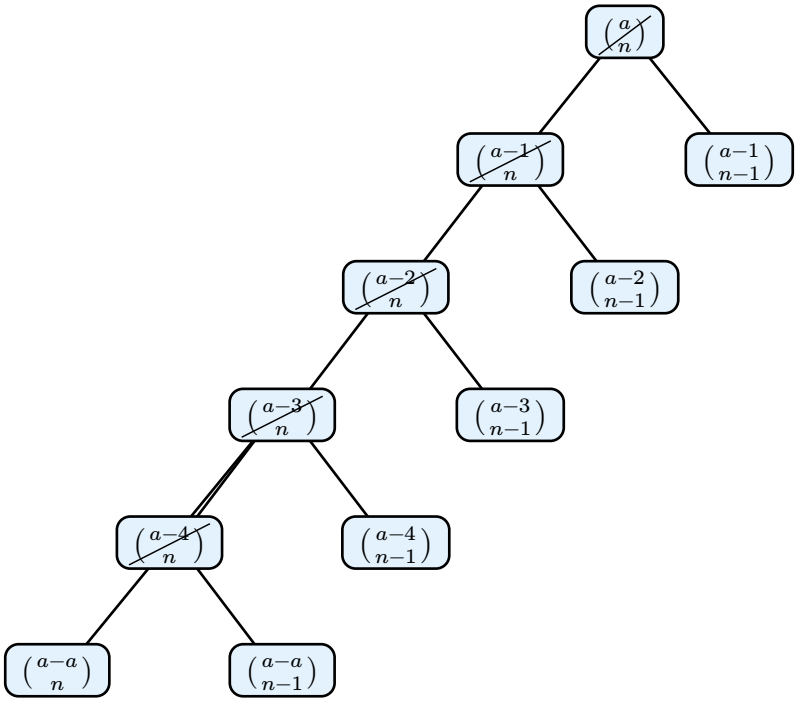
let the product and sum of these outer binom terms grouped by sum level (n), $= D_{r,n}$

$$x^r = D_{r,1}\sum_{b=0}^{x-1}[b^0] + D_{r,2}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[c^0] + D_{r,3}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[d^0] + ... + D_{r,r}\overbrace{\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}...\sum_{n_n=0}^{n_{n-1}-1}[n_n^0]}^{r\text{ sums}} + 0^r$$

pascals identity

$$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$$

apply pascals identity recursivley to lhs



$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\binom{b}{n-1}\right] + \binom{0}{n}$$

apply the identity to the inner $\binom{b}{n-1}$

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\binom{c}{n-2}\right] + \binom{0}{n-1}\right] + \binom{0}{n}$$

expanding the inner term n-1 times, for n total sums, causes its inner term to be 1, as $\binom{a \geq 0}{0} = 1$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\dots \sum_{n_n=0}^{n_{n-1}-1}\left[\binom{n_n}{n-n}\right] + \binom{0}{n-(n-1)}\dots\right]\binom{0}{n-2}\right] + \binom{0}{n-1}\right]}_{\text{n sums}} + \binom{0}{n}$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\dots \sum_{n_n=0}^{n_{n-1}-1}\left[\binom{n_n}{0}\right] + \binom{0}{n-(n-1)}\dots\right]\binom{0}{n-2}\right] + \binom{0}{n-1}\right]}_{\text{n sums}} + \binom{0}{n}$$

as we only expanded n-1 times the $\binom{0}{n_{n-1}}$ only reach $\binom{0}{n-(n-1)} = \binom{0}{1} = 0$, as $\binom{0}{n>0} = 0$

if $n = 0$, the sums arent expanded and only the $\binom{0}{n}$ remains

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\dots \sum_{n_n=0}^{n_{n-1}-1}[1]\right]\right]\right]}_{\text{n sums}} + \binom{0}{n}$$

the sums in the binomial therom expansion and pascals identity expansions have the same bounds change

$$x^r = D_{r,1}\sum_{b=0}^{x-1}[1] + D_{r,2}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}[1] + D_{r,3}\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}[1] + ... + D_{r,r}\overbrace{\sum_{b=0}^{x-1}\sum_{c=0}^{b-1}\sum_{d=0}^{c-1}...\sum_{n_n=0}^{n_{n-1}-1}}^{r\text{ sums}}[1] + 0^r$$

$$x^r = D_{r,1}\binom{x}{1} + D_{r,2}\binom{x}{2} + D_{r,3}\binom{x}{3} + ... + D_{r,r}\overbrace{\binom{x}{r}}^{r\text{ sums}} + 0^r$$

$$x^r = \sum_{b=1}^{r}\left[D_{r,b}\binom{x}{b}\right] + 0^r$$

$$\text{let } D_{r,0} = 0^r$$

$$x^r = \sum_{b=0}^{r}\left[D_{r,b}\binom{x}{b}\right]$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

every expansion of b^c adds a sum layer to the expanded terms, the $\sum_{b=0}^{x-1}$ is this accumulator

each new layer of b_n^c adds a sum to the expanded terms, so each new layer can only produce diffrences upto n (beacuse it has a term with a minimum of n sums)

therfore to get the nth diffrence you would on need to expand n times, (n-1 times if you count the base identity as the 0th expansion beacuse it adds a outer sum to the terms); diliberate expansion of terms with r

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}b^0 + \binom{r}{1}b^1 + \binom{r}{2}b^2...\binom{r}{r-1}b^{r-1}\right] + 0^r$$

$$b^0 = 1$$

$$b^1 = \sum_{c=0}^{x-1}\left[\overbrace{\binom{1}{0}c^0}^{1}\right]$$

$$b^2 = \sum_{c=0}^{x-1}\left[\overbrace{\binom{2}{0}c^0}^{1} + \binom{2}{1}\sum_{d=0}^{c-1}\left[\overbrace{\binom{1}{0}d^0}^{1}\right]\right]$$

each layer contains a 1 term it will have L (layer) amount of sums (for $r \geq 0$)

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=1}^{r-1}\left[\binom{r}{c}b^c\right] + \overbrace{\overbrace{\binom{r}{0}b^0}^{1\text{ at }r\geq 1}}\right] + \overbrace{0^r}^{1\text{ at }r=0}$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}b^c\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{r}\left[D_{r,b}\binom{x}{b}\right]$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}\sum_{d=0}^{c}\left[D_{c,d}\binom{b}{d}\right]\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}\sum_{d=0}^{0}\left[D_{0,d}\binom{b}{d}\right] + \binom{r}{1}\sum_{d=0}^{1}\left[D_{1,d}\binom{b}{d}\right] + ... + \binom{r}{r-1}\sum_{d=0}^{r-1}\left[D_{r-1,d}\binom{b}{d}\right]\right] + 0^r$$

grouping the terms by $\binom{b}{d}$ tells the coefficents of each diffrence level, this will produce a formula for $D_{r,c}$ in terms of $D_{r,c}$ from lower degree terms.

since every time you lower the degree you loose a diffrence level until D_0, there will be a falling number of each diffrence level in the sum. $\sum^{r-1} D_0 + \sum^{r-2} D_1 + ... \sum^{r-r} D_{r-1}$ (grouping the inner sum, excluding $\sum^{x-1}$)

$$x^r = \sum_{b=0}^{x-1}\left[\binom{r}{0}\sum_{d=0}^{0}\left[D_{0,d}\binom{b}{d}\right] + \binom{r}{1}\sum_{d=0}^{1}\left[D_{1,d}\binom{b}{d}\right] + ... + \binom{r}{r-1}\sum_{d=0}^{r-1}\left[D_{r-1,d}\binom{b}{d}\right]\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\overbrace{\left[\binom{r}{0}D_{0,0} + \binom{r}{1}D_{1,0} + ... + \binom{r}{r-1}D_{r-1,0}\right]}^{D_{r,0+1}}\binom{b}{0} + ... + \overbrace{\left[\binom{r}{r-1}D_{r-1,r-1}\right]}^{D_{r,r-1+1}}\binom{b}{r-1}\right] + 0^r$$

beacuse of outer sum the diffrence level has 1 added to it,the grouped coefficents on $\binom{b}{n}$ tells the diffrence level of the nth diffrence as this would expand to n falling sums

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[D_{c,0}\binom{r}{c}\right]\binom{b}{0} + \sum_{c=1}^{r-1}\left[D_{c,1}\binom{r}{c}\right]\binom{b}{1} + \ldots + \sum_{c=r-1}^{r-1}\left[D_{c,r-1}\binom{r}{c}\right]\binom{b}{r-1}\right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1}\left[\sum_{c=0}^{r-1}\left[\binom{b}{c}\sum_{d=c}^{r-1}\left[D_{d,c}\binom{r}{d}\right]\right]\right] + 0^r$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[\sum_{d=0}^{c-1}\left[\ldots\sum_{n_n=0}^{n_{n-1}-1}[1]\right]\right]\right]}_{\text{n sums}} + \binom{0}{n}$$

$$x^r = \sum_{c=0}^{r-1}\left[\overbrace{\binom{x}{c+1}}^{\text{Diffrence sums (c+1)}}\overbrace{\sum_{d=c}^{r-1}\left[D_{d,c}\binom{r}{d}\right]}^{\text{Diffrence coefficents}}\right] + \overbrace{0^r}^{\text{0th Diffrence}}$$

$$D_{r,0} = 0^r$$

$$D_{0,0} = 1$$

$$D_{r,c+1} = \overbrace{\sum_{d=c}^{r-1}\left[D_{d,c}\binom{r}{d}\right]}^{\text{Diffrence coefficents}}$$

$$D_{r,c} = \overbrace{\sum_{d=c-1}^{r-1}\left[D_{d,c-1}\binom{r}{d}\right]}^{\text{Diffrence coefficents}}$$

$$D_{r,c} = \sum_{d=c-1}^{r-1}\left[D_{d,c-1}\binom{r}{d}\right]$$

$$D_{r,c} = \sum_{d=c-1}^{r-1}\left[\sum_{e=c-2}^{d-1}\left[D_{e,c-2}\binom{d}{e}\right]\binom{r}{d}\right]$$

expand c-1 times, (c times if you count the base diffrences identity as the 0th expansion)

$$D_{r,c} = \overbrace{\sum_{d=c-1}^{r-1}\left[\sum_{e=c-2}^{d-1}\left[\sum_{f=c-3}^{e-1}\left[\ldots\sum_{n_{c-1}=c-c}^{n_{c-2}-1}\left[D_{n_{c-1},c-c}\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c\text{ sums}}$$

$$D_{r,c} = \sum_{d=c-1}^{r-1} \left[D_{d,c-1} \binom{r}{d} \right]$$

since x^r only has diffrences upto level r , $D_{r,(c>r)}$ can be treated as 0

$$D_{r,c} = \sum_{d=0}^{r-1} \left[D_{d,c-1} \binom{r}{d} \right]$$

$$D_{r,c} = \sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[D_{e,c-2} \binom{d}{e} \right] \binom{r}{d} \right]$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\dots \sum_{n_{c-1}=0}^{n_{c-2}-1} \left[D_{n_{c-1},c-c} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\dots \sum_{n_{c-1}=0}^{n_{c-2}-1} \left[D_{n_{c-1},0} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

$$D_{r,0} = 0^r$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1} \left[\sum_{e=0}^{d-1} \left[\sum_{f=0}^{e-1} \left[\dots \sum_{n_{c-1}=0}^{n_{c-2}-1} \left[0^{n_{c-1}} \binom{n_{c-2}}{n_{c-1}} \right] \dots \binom{e}{f} \right] \binom{d}{e} \right] \binom{r}{d} \right]}^{c \text{ sums}}$$

since $0^r = 1$ only at $r = 0$, the inner sums $0^{n_{c-1}}$ will only be 1 at the start, since $0^{\neq 0}$ is 0 the $\binom{n_{c-2}}{n_{c-1}}$ will cancle out when not at the start of the sum, the sum only produces a single non zero run at its start, as long as the inner most upper bound is not negative, if $r \geq c$ the inner sum is guarnted to run at least once

the inner sum only runs once if upper bound is not negative

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}-1}^{n_{c-2}-1}[1]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\sum_{n_{c-2}-1}^{n_{c-2}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c \text{ sums}}$$

for the inner most sum to run the outer sums lower bound must be at least 1, $n_{c-2} \geq 1$, if it is 0 it will not run

$1 - 0^{n_{c-2}}$ is only 1 if $n_{c-2} \neq 0$ or $n_{c-2} > 0$ and $n_{c-2} < 0$

since n_{c-2} starts at 0 and goes upto $n_{c-3} - 1$ it can be negative or its parent sum wouldnt have run it

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[[1-0^{n_{c-2}}]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right)]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[[-0^{n_{c-2}}]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right)]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[[-0^{n_{c-2}}]\binom{n_{c-3}}{n_{c-2}}\right]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$-0^{n_{c-2}}$ is only -1 if n_{c-2} is 0, the base of the sum, $\binom{n_{c-3}}{n_{c-2}}$ is 1 at the base

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-2}=0}^{n_{c-3}-1}[1]\binom{n_{c-3}}{n_{c-2}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right)]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\sum_{n_{c-3}-1}^{n_{c-3}-1}[-1]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}}$$

$$\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\sum_{n_{c-3}-1}^{n_{c-3}-1}[-1]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-1 \text{ sums}} =$$

$$\overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\dots\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[[-1+0^{n_{c-3}}]\binom{n_{c-4}}{n_{c-3}}\right]\dots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c-2 \text{ sums}}$$

$\sum_{b=0}^{a}[0^b]$ will produce 1 if $a \geq 0$, it will produce 0 if $a < 0$ it is a switch that is only on if the upper sum bound is ≥ 0
$\sum_{b=0}^{a}[0^b] = a \geq 0 = a == 0 \mid a > 0$ it defaults to 1 only if $a < 0$ will it be 0, defualt - (1 if less than 0) = 1 - (1 if less than 0)

$$\sum_{b=0}^{a}[0^b] = 1 - \begin{cases} 1 \text{ if } a < 0 \\ 0 \text{ if } a == 0 \\ 0 \text{ if } a > 0 \end{cases} = \begin{cases} 0 \text{ if } a < 0 \\ 1 \text{ if } a == 0 \\ 1 \text{ if } a > 0 \end{cases}$$

$$0^a = \begin{cases} 1 \text{ if } a == 0 \\ 0 \text{ or undefined if } a < 0 \\ 0 \text{ if } a > 0 \end{cases}$$

$$0^{a+1} = \begin{cases} 1 \text{ if } a == (-1) \\ 0 \text{ or undefined if } a < (-1) \\ 0 \text{ if } a > (-1) \end{cases}$$

$1 - 0^{a+1}$ will only be zero if $a = -1$, it is also the inverts 0^{a+1} as it defualts to 1 and is only 0 when $0^{a+1} == 1$

$$1 - 0^{a+1} = \begin{cases} 0 \text{ if } a == (-1) \\ 1 \text{ or undefined if } a < (-1) \\ 1 \text{ if } a > (-1) \end{cases} = \begin{cases} 0 \text{ if } a == (-1) \\ 1 \text{ or undefined if } a < (-1) \\ 1 \text{ if } a == (0) \\ 1 \text{ if } a > (0) \end{cases}$$

2 of the 4 cases match $\begin{cases} 0 \text{ if } a < 0 \\ 1 \text{ if } a == 0 \text{, the 1st case is paritally matched (only for } a == -1 \text{ not for } a < (-1)) \\ 1 \text{ if } a > 0 \end{cases}$

$$\forall a \text{ such that } a == -1 \mid a > 0 \mid a == 0 : \sum_{b=0}^{a}[0^b] = 1 - 0^{a+1}$$

$$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}[0^b] = 1 - 0^a$$

$$\text{let } k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots \sum_{n_n=0}^{n_{n-1}-1}[0_n^n]\ldots\right]\right]\right]}^{n \text{ sums}}$$

$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}[0^b] = 1 - 0^a$, the lower bounds at every sum layer in k have a lower bound of 0, since the upper bound of the next sum is the incrementer-1 and since this incrementer never drops below 0 the identity could be applied to every layer of k. since the identity produces 0^a terms inside a nested sum those terms can have the identity recursivley applied to themselves.The final sum cant be brough out as n_0 is not guranted to to be ≥ 0

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[\sum_{n_{n-1}=0}^{n_{n-2}-1}\left[\sum_{n_n=0}^{n_{n-1}-1}[0_n^n]\right]\right]...\right]\right]\right]}^{n \text{ sums}}$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[\sum_{n_{n-1}=0}^{n_{n-2}-1}[1-0^{n_{n-1}}]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[-(1-0^{n_{n-2}})+\sum_{n_{n-1}=0}^{n_{n-2}-1}[1]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

every expansions inverts the sign of the identity

$$1-0^a = (-1)^0 + (-1)^1 0^a$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[(-1)^1\left((-1)^0+(-1)^1 0^{n_{n-2}}\right)+\sum_{n_{n-1}=0}^{n_{n-2}-1}[(-1)^0]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

LHS and RHS are repeatedly multipled by $(-1)^1$, RHS starts with a degree higher (-1) coefficent

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[(-1)^1+\sum_{n_{n-1}=0}^{n_{n-2}-1}[(-1)^0]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

$$k = \overbrace{\sum_{n_1=0}^{n_0-1}\left[(-1)^{n-(2)}+(-1)^{n-(2)+1}0^{n_1}+\sum_{n_2=0}^{n_1-1}\left[(-1)^{n-(3)}+\sum_{n_3=0}^{n_2-1}\left[(-1)^{n-(4)}+...\sum_{n_{n-2}=0}^{n_{n-3}-1}\left[(-1)^{n-(n-1)}+\sum_{n_{n-1}=0}^{n_{n-2}-1}[(-1)^{n-n}]\right]...\right]\right]\right]}^{n-1 \text{ sums}}$$

$$k = (-1)^{n-(n)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-1}=0}^{n_{n-2}-1}[1]\right]\right]}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}[1]\right]\right]}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\sum_{n_1=0}^{n_0-1}[(1+(-1)^1 0^{n_1})]}^{1 \text{ sums}}$$

$$= (-1)^{n-(n)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-1}=0}^{n_{n-2}-1}[1]\right]\right]}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[...\sum_{n_{n-2}=0}^{n_{n-3}-1}[1]\right]\right]}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\sum_{n_1=0}^{n_0-1}[1]}^{1 \text{ sums}} + (-1)^{n-(2)+1}\overbrace{\sum_{n_1=0}^{n_0-1}[0^{n_1}]}^{1 \text{ sums}}$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(2)+1}\overbrace{\sum_{n_1=0}^{n_0-1}[0^{n_1}]}^{1 \text{ sums}}$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(2)+1}\sum_{n_0-1}^{n_0-1}\left[\underbrace{\overbrace{\binom{n_0}{0}}^{0 \text{ sums}}}_{1} - 0^{n_0}\right]$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(1)}\sum_{n_0-1}^{n_0-1}\left[\underbrace{\overbrace{\binom{n_0}{0}}^{0 \text{ sums}}}_{1}\right] + (-1)^n\sum_{n_0-1}^{n_0-1}[0^{n_0}]$$

$$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}[0^b] = 1-0^a$$

$$\forall n_0 \in \mathbb{Z}, n_0 \geq 0 \Rightarrow$$

$$k = (-1)^{n-(n)}\overbrace{\binom{n_0}{n-1}}^{n-1 \text{ sums}} + (-1)^{n-(n-1)}\overbrace{\binom{n_0}{n-2}}^{n-2 \text{ sums}} + ... + (-1)^{n-(2)}\overbrace{\binom{n_0}{1}}^{1 \text{ sums}} + (-1)^{n-(1)}\underbrace{\overbrace{\binom{n_0}{0}}^{0 \text{ sums}}}_{1} + (-1)^n 0^{n_0}$$

$$\forall n_0 \in \mathbb{Z}, n_0 \geq 0 \Rightarrow k = \sum_{b=0}^{n-1}\left[(-1)^b\binom{n_0}{n-(b+1)}\right] + (-1)^n 0^{n_0}$$

$$\forall n_0 \in \mathbb{Z}, n_0 \geq 0 \Rightarrow k = \sum_{b=0}^{n-1}\left[(-1)^{n-(b+1)}\binom{n_0}{b}\right] + (-1)^n 0^{n_0}$$

$$D_{r,c} = \overbrace{\sum_{d=0}^{r-1}\left[\sum_{e=0}^{d-1}\left[\sum_{f=0}^{e-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[0^{n_{c-1}}\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{e}{f}\right]\binom{d}{e}\right]\binom{r}{d}\right]}^{c \text{ sums}}$$

$$\binom{a}{0} = 1$$

only when $b = 0$ does $0^b = 1$, so all other $\binom{a}{b}$ besides the base $\binom{a}{0}$, vanish

$$\forall a \in \mathbb{Z}, a \geq 0 \Rightarrow \sum_{b=0}^{a-1}\left[0^b\binom{a}{b}\right] = (1-0^a)\binom{a}{0} = 1 - 0^a$$

this sum is identitcal to k besides each layer of a sum of 1 has binom coefficents

let $n_0 = r$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_c=0}^{n_{c-1}-1}\left[0^{n_c}\binom{n_{c-1}}{n_c}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\sum_{n_c=0}^{n_{c-1}-1}\left[0^{n_c}\binom{n_{c-1}}{n_c}\right]\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[(1-0^{n_{c-1}})\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-1 \text{ sums}}$$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[(-1)^{c-c}\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-1 \text{ sums}} +$$

$$\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[(-1)^{c-(c-1)}\binom{n_{c-3}}{n_{c-2}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-2 \text{ sums}}$$

$$+\ldots+\overbrace{\sum_{n_1=0}^{n_0-1}\left[((-1)^{c-2}+(-1)^{c-2+1}0^{n_1})\binom{n_0}{n_1}\right]}^{1 \text{ sums}}$$

$$D_{r,c} = (-1)^{c-c}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-1 \text{ sums}} +$$

$$(-1)^{c-(c-1)}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\ldots\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\binom{n_{c-3}}{n_{c-2}}\right]\ldots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c-2 \text{ sums}}$$

$$+\ldots+(-1)^{c-2}\overbrace{\sum_{n_1=0}^{n_0-1}\left[\binom{n_0}{n_1}\right]}^{1 \text{ sums}} + (-1)^{c-1}\overbrace{\sum_{n_1=0}^{n_0-1}\left[0^{n_1}\binom{n_0}{n_1}\right]}^{1 \text{ sums}}$$

$$\text{binomial therom: } (x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

if both x and $a = 1$ then only the sum of binomial terms will remain as $1^n = 1$

$$(1+1)^r = \sum_{b=0}^{r}\left[\binom{r}{b}(1)(1)\right]$$

$$2^r = \sum_{b=0}^{r}\left[\binom{r}{b}\right]$$

$$2^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}\right] + \overbrace{\binom{r}{r}}^{1}$$

$$2^r - 1 = \sum_{b=0}^{r-1}\left[\binom{r}{b}\right]$$

$$\text{let } l = \sum_{b=0}^{a-1}\left[\binom{a}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}\right]\right] = \sum_{b=0}^{a-1}\left[\binom{a}{b}(2^b-1)\right] = \sum_{b=0}^{a-1}\left[\binom{a}{b}(-1) + \binom{a}{b}(2^b)\right] = (-1)\sum_{b=0}^{a-1}\left[\binom{a}{b}\right] + \sum_{b=0}^{a-1}\left[\binom{a}{b}(2^b)\right]$$

$$\text{let } a' = 1, \text{let } x' = 2$$

$$(1+2)^r = 3^r = \sum_{b=0}^{r}\left[\binom{r}{b}2^b(1)\right]$$

$$3^r - 1(2^r) = \sum_{b=0}^{r-1}\left[\binom{r}{b}2^b(1)\right]$$

$$l = (-1)\sum_{b=0}^{a-1}\left[\binom{a}{b}\right] + 3^a - 2^a = (-1)(2^a-1) + 3^a - 2^a$$

$$(a+b)^r = \sum_{s=0}^{r}\left[\binom{r}{s}a^s b^{r-s}\right]$$

$$\text{let } b = 1$$

$$(a+1)^r = \sum_{s=0}^{r}\left[\binom{r}{s}a^s(1)\right]$$

$$\text{let } a = c+1$$

$$(a+1)^r = \sum_{s=0}^{r}\left[\binom{r}{s}(c+1)^s(1)\right] = \sum_{s=0}^{r}\left[\binom{r}{s}\sum_{s'=0}^{s}\left[\binom{s}{s'}(c)^{s'}\right]\right]$$

this strucutre is almost identical to the diffrences strucutre, the upper bound is the only diffrence, for this to converge to $0^{s''''\cdots}$ the max upper bound must decrease by -1 at each layer, if it does the base of the power term will reach 0. facotr out last sum term to get falling upper bounds.

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-1)^b(1)\right]$$

if x=2 the sum collapses to sum of binomials, if x=1 the sum collapse to a sum of 0^b

apply the identity to each x, here x‘ = x-1

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}[\binom{r}{b}(((x-2)+1)^b(1)]$$

$$((x-1)+1)^r = x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}(x-2)^c\right]\right]$$

if x=3 the sum collapses to a double sum of binomials, if x=2 the sum collapses to a double sum of 0^c

every expansion increase the number of sums by 1 and decreases the value of the inner x‘ by 1

$$((x-1)+1)^r = x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\right]...\sum_{n_c=0}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]\right]}^{c\text{ sums}}$$

we can pull out the last term in each sum to get falling sums

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left(\sum_{c=0}^{b-1}\left[\binom{b}{c}\left(...\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]+\underbrace{\binom{n_{c-1}}{n_{c-1}}}_{1}(x-c)^{n_{c-1}}\right]...\right)+\underbrace{\binom{b}{b}}_{1}(x-2)^b\right)\right]}^{c\text{ sums}}+\underbrace{\binom{r}{r}}_{1}(x-1)^r$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left(\sum_{c=0}^{b-1}\left[\binom{b}{c}\left(...\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]+(x-c)^{n_{c-1}}\right]...\right)+(x-2)^b\right)\right]}^{c\text{ sums}}+(x-1)^r$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}(x-c)^{n_c}\right]\right]...\right]}^{c\text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}(x-c)^{n_{c-1}}\right]\right]...\right]}^{c-1\text{ sums}}+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\binom{n_{c-3}}{n_{c-2}}(x-(c-1))^{n_{c-2}}\right]\right]...\right]}^{c-2\text{ sums}}$$

$$+...+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-2)^b\right]}^{1\text{ sums}}+(x-1)^r$$

i can use a (x-1)^r to absorb lower sums

$$(x-1)^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}((x-1)-(c-1))^{n_{c-1}}\right]\right]...\right]}^{c-1\text{ sums}}$$

$$\overbrace{\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\binom{n_{c-3}}{n_{c-2}}((x-1)-(c-1))^{n_{c-2}}\right]\right]...\right]}^{c-2\text{ sums}}+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\binom{n_{c-4}}{n_{c-3}}((x-1)-(c-2))^{n_{c-3}}\right]\right.\right.}^{c-3\text{ sums}}$$

$$+...+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}((x-1)-2)^b\right]}^{1\text{ sums}}+((x-1)-1)^r$$

$$(x-1)^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}(x-c)^{n_{c-1}}\right]\right]...\right]}^{c-1 \text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-2}=0}^{n_{c-3}-1}\left[\binom{n_{c-3}}{n_{c-2}}(x-c)^{n_{c-2}}\right]\right]...\right]}^{c-2 \text{ sums}} + \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}...\sum_{n_{c-3}=0}^{n_{c-4}-1}\left[\binom{n_{c-4}}{n_{c-3}}(x-(c-1))^{n_{c-3}}\right]\right]...\right]}^{c-3 \text{ sums}}$$

$$+... + \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}(x-3)^b\right]}^{1 \text{ sums}} + (x-2)^r$$

subtracting these would result in ony 1 overlaping term (c-1 sums) being removed and it would add more terms to the tail,

only the highest sums term would be removed,

this will work but it requires accumulating the overlap from the previous powers subtracted

if we could simplify the lower sums into closed form binomial-power terms we can subtract them. if x=c the c nested sums matches $D_{r,c}$ and the lower sums can be factored as a sum of 1 with binom coeffcients at lower sum layers.

$$\text{let } m = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\dots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\sum_{n_c=0}^{n_{c-1}-1}\left[1\binom{n_{c-1}}{n_c}\right]\binom{n_{c-2}}{n_{c-1}}\right]\dots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

$$(1+1)^r = \sum_{b=0}^{r}\left[\binom{r}{b}\right]$$

$$(1+1)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}\right] - \binom{r}{r}$$

$$(2)^r - 1 = \sum_{b=0}^{r-1}\left[\binom{r}{b}\right]$$

$$\sum_{b=0}^{r-1}\left[\binom{r}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}\right]\right] = \sum_{b=0}^{r-1}\left[\binom{r}{b}((2)^b - 1)\right] = \sum_{b=0}^{r-1}\left[\binom{r}{b}(-1)\right] + \sum_{b=0}^{r-1}\left[\binom{r}{b}((2)^b)\right]$$

$$(2+1)^r - \binom{r}{r}2^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}2^b\right]$$

coefficent match, subtract last term, split sum, repeat

$$\text{let } x = \sum_{b=0}^{a-1}\left[\binom{a}{b}\sum_{c=0}^{b-1}\left[\binom{b}{c}m'^c\right]\right]$$

coefficent match and subtract creating 2 terms

$$(m'+1)^b - \binom{b}{b}m'^b = \sum_{c=0}^{b-1}\left[\binom{b}{c}m'^c\right] = (m'+1)^b - m'^b = \sum_{c=0}^{b-1}\left[\binom{b}{c}m'^c\right]x = \sum_{b=0}^{a-1}\left[\binom{a}{b}\left((m'+1)^b - m'^b\right)\right]$$

split the 2 terms

$$x = \sum_{b=0}^{a-1}\left[\binom{a}{b}(-m'^b)\right] + \sum_{b=0}^{a-1}\left[\binom{a}{b}\left((m'+1)^b\right)\right]$$

repeat this process for the 2 new sums

$$\sum_{b=0}^{a-1}\left[\binom{a}{b}(-m'^b)\right] = ((-m')+1)^a - (-m')^a$$

$$\sum_{b=0}^{a-1}\left[\binom{a}{b}\left((m'+1)^b\right)\right] = ((m'+1)+1)^a - (m'+1)^a$$

$$x = ((-m')+1)^a - (-m')^a + ((m'+1)+1)^a - (m'+1)^a$$

$$\text{goal: } \overbrace{\sum_{b=0}^{a}\binom{a}{b}\sum_{c=0}^{b}\binom{b}{c}\sum_{d=0}^{c}\binom{c}{d}\ldots\sum_{n_m=0}^{n_{m-1}}\binom{n_{m-1}}{n_m}}^{m \text{ sums}}$$

$$(x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

rename: $a \to m, r \to a$

$$(x+m)^r = \sum_{b=0}^{a}\left[\binom{a}{b}x^b m^{a-b}\right] \text{ let } m = 1$$

$$(x+1)^r = \sum_{b=0}^{a}\left[\binom{a}{b}x^b\right]$$

the sums at higher depths have the same structure as this

if that inner x^b = (x+1)^b it would recurse into another sum like the identity states

$$((x+1)+1)^r = \sum_{b=0}^{a}\left[\binom{a}{b}(x+1)^b\right] = \sum_{b=0}^{a}\left[\binom{a}{b}\sum_{c=0}^{b}\left[\binom{b}{c}x^b\right]\right]$$

every +1 can be stripped away by the binomial therom into a sum

$$m = \overbrace{1+1+1+1+1+\ldots+1}^{m \text{ 1's}}$$

$$(x+m)^a = \overbrace{\sum_{b=0}^{a}\binom{a}{b}\sum_{c=0}^{b}\binom{b}{c}\sum_{d=0}^{c}\binom{c}{d}\ldots\sum_{n_m=0}^{n_{m-1}}\left[\binom{n_{m-1}}{n_m}x^{n_m}\right]}^{m \text{ sums}}$$

the goal has been achived, i can even control the inermost power too

a neat symetrey beacuse of the commutitive property of addition:

$$(x+m)^a = (m+x)^a = \overbrace{\sum_{b=0}^{a}\binom{a}{b}\sum_{c=0}^{b}\binom{b}{c}\sum_{d=0}^{c}\binom{c}{d}\ldots\sum_{n_m=0}^{n_{m-1}}\left[\binom{n_{m-1}}{n_m}x^{n_m}\right]}^{m \text{ sums}} = \overbrace{\sum_{b=0}^{a}\binom{a}{b}\sum_{c=0}^{b}\binom{b}{c}\sum_{d=0}^{c}\binom{c}{d}\ldots\sum_{n_x=0}^{n_{x-1}}\left[\binom{n_{x-1}}{n_x}m^{n_x}\right]}^{x \text{ sums}}$$

let $n_0 = r$

$$D_{r,c} = \overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\dots\sum_{n_c=0}^{n_{c-1}-1}\left[0^{n_c}\binom{n_{c-1}}{n_c}\right]\dots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

goal: $$\overbrace{\sum_{n_1=0}^{n_0-1}\left[\sum_{n_2=0}^{n_1-1}\left[\sum_{n_3=0}^{n_2-1}\left[\dots\sum_{n_c=0}^{n_{c-1}-1}\left[0^{n_c}\binom{n_{c-1}}{n_c}\right]\dots\binom{n_2}{n_3}\right]\binom{n_1}{n_2}\right]\binom{n_0}{n_1}\right]}^{c \text{ sums}}$$

the identity we dervied gets use very close already

$$(0+c)^a = \overbrace{\sum_{b=0}^{a}\binom{a}{b}\sum_{c=0}^{b}\binom{b}{c}\sum_{d=0}^{c}\binom{c}{d}\dots\sum_{n_c=0}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}0^{n_c}\right]}^{c \text{ sums}}$$

remeber from our origional derivation:

$$((x+1)+1)^r = \sum_{b=0}^{a}\left[\binom{a}{b}(x+1)^b\right] = \sum_{b=0}^{a}\left[\binom{a}{b}\sum_{c=0}^{b}\left[\binom{b}{c}x^b\right]\right]$$

every +1 can be stripped away by the binomial therom into a sum

using the +1 stripping rule

the sum can be decomposed into sums of (0+d)^(n_d) where d is depth and n_d is the incrementer at that depth

$$(0+c)^a = \sum_{b=0}^{a}\binom{a}{b}\sum_{c=0}^{b}\binom{b}{c}\sum_{d=0}^{c}\binom{c}{d}\dots\overbrace{\sum_{n_c=0}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}0^{n_c}\right]}^{\overbrace{\overbrace{\overbrace{\overbrace{(0+c)^a}^{c \text{ sums}}}^{(0+(c-1))^b}}^{\left(0+(c-2)^{c'}\right)}}^{(0+c-c)^{n_c}}}$$

just attach the trailing sums to these terms and you can strip the last term from each sum

$$(0+c)^a = \overbrace{\overbrace{\overbrace{\overbrace{(0+c)^a}^{(0+(c-1))^b}}^{\left(0+(c-2)^{c'}\right)}}^{(0+c-c)^{n_c}}}^{c \text{ sums, (D_(r,c))}} \sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots\overbrace{\sum_{n_c=0}^{n_{c-1}-1}\left[\binom{n_{c-1}}{n_c}0^{n_c}\right]}$$

$$+\overbrace{\sum_{b=a}^{a}\binom{a}{b}\overbrace{(0+(c-1))^b}^{(0+c)^a}}$$

$$+\overbrace{\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=b}^{b}\binom{b}{c}\overbrace{\left(0+(c-2)^{c'}\right)}^{(0+(c-1))^b}}^{(0+c)^a}$$

$$+\ldots+$$

$$+\overbrace{\overbrace{\overbrace{\overbrace{(0+c)^a}^{(0+(c-1))^b}}^{\left(0+(c-2)^{c'}\right)}}^{(0+c-c)^{n_c}}}^{c \text{ sums}} \sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots\overbrace{\sum_{n_c=n_{c-1}}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}0^{n_c}\right]}$$

now if we move these end terms to the otherside of the equation we have another formula for D_(r,c)

$$D_{r,c} = (0+c)^a$$

$$-\overbrace{\sum_{b=a}^{a}\binom{a}{b}\overbrace{(0+(c-1))^b}^{(0+c)^a}}$$

$$-\overbrace{\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=b}^{b}\binom{b}{c}\overbrace{\left(0+(c-2)^{c'}\right)}^{(0+(c-1))^b}}^{(0+c)^a}$$

$$-\ldots-$$

$$-\overbrace{\overbrace{\overbrace{\overbrace{(0+c)^a}^{(0+(c-1))^b}}^{\left(0+(c-2)^{c'}\right)}}^{(0+c-c)^{n_c}}}^{c \text{ sums}} \sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots\overbrace{\sum_{n_c=n_{c-1}}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}0^{n_c}\right]}$$

lets clean this up

first lets remove the overbrace

$$D_{r,c} = (0+c)^a$$

$$-\sum_{b=a}^{a}\binom{a}{b}(0+(c-1))^b$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=b}^{b}\binom{b}{c}\left(0+(c-2)^{c'}\right)$$

$$-\ldots-$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots\sum_{n_c=n_{c-1}}^{n_{c-1}}\left[\binom{n_{c-1}}{n_c}0^{n_c}\right]$$

second lets evaluate the innermost sums

$$D_{r,c} = (0+c)^a$$

$$-\binom{a}{a}(0+(c-1))^a$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\binom{b}{b}(0+(c-2)^b)$$

$$-\ldots-$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots\left[\binom{n_{c-1}}{n_{c-1}}0^{n_{c-1}}\right]$$

thrid lets remove the 0's

$$D_{r,c} = (c)^a$$

$$-\binom{a}{a}(c-1)^a$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\binom{b}{b}(c-2)^b$$

$$-\ldots-$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots\left[\binom{n_{c-1}}{n_{c-1}}0^{n_{c-1}}\right]$$

fourth lets remove the binom(a,a)'s

$$D_{r,c} = (c)^a$$

$$-(c-1)^a$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}(c-2)^b$$

$$-\ldots-$$

$$-\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\ldots[0^{n_{c-1}}]$$

$$D_{r,c} = (c)^a$$
$$-(c-1)^a$$
$$-\sum_{b=0}^{a-1}\binom{a}{b}(c-2)^b$$
$$-\dots-$$
$$-\overbrace{\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c=0}^{b-1}\binom{b}{c}\sum_{d=0}^{c-1}\binom{c}{d}\dots\sum_{n_{c-1}=0}^{n_{c-2}-1}\left[\binom{n_{c-2}}{n_{c-1}}0^{n_{c-1}}\right]}^{c-1\text{ sums}}$$

notice the last term is just a nested sum of 0^(last_incrementer) with binom terms at each layer

we can apply the 0 sum formula of D_(r,c) to get that term into closed form

in this case it would simplify to D_(r,c-1) (a=r) rename r->a and apply the substition

$$D_{a,c} = (c)^a$$
$$-(c-1)^a$$
$$-\sum_{b=0}^{a-1}\binom{a}{b}(c-2)^b$$
$$-\dots-\overbrace{\sum_{b=0}^{a-1}\binom{a}{b}\sum_{c'=0}^{b-1}\binom{b}{c'}\dots\sum_{n_c-2=0}^{n_{c-3}-1}\binom{n_{c-3}}{n_{c-2}}(c-3)^b}^{c-2\text{ sums}}$$
$$-\overbrace{D_{a,c-1}}^{c-1\text{ sums}}$$

if we apply this formula to itself we get a staircase like structure

by this i mean each expansion removes the last term and adds 1 term to each lower term, like a stairscase